

# Excel Shrink: High-Performance Bytestream Parsing and Cleaning of Bloated Excel Spreadsheets

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**Abstract**—This paper introduces Excel Shrink, a tool designed to mitigate the pervasive problem of file bloat in Microsoft Excel spreadsheets. By analyzing Excel’s Office Open XML structure, we identify common sources of file bloat—such as cells, rows and columns that are empty, or have a style assigned that is invisible to the user. We propose a novel, streaming-based, regex-driven approach to reduce file size without altering the visual appearance. Our method employs a state-machine to manage the hierarchy of XML documents without relying on conventional XML parsing techniques by processing XML data incrementally at the bytestream level, and it leverages multithreading for performance improvements. This enables Excel Shrink to outperform Microsoft Excel when loading bloated files, and cleaning plus opening the shrunk file is faster than opening the bloated file directly. Evaluation on large-scale real-world spreadsheets from the German Federal Statistical Office and data.gov demonstrates that Excel Shrink can achieve file size reductions of up to 99.99% and dramatically improve file loading times. This work not only provides practical insights for developers and end users but also opens new avenues for the efficient management of large-scale semi-structured data in spreadsheet environments.

**Index Terms**—component, formatting, style, styling, insert.

## I. INTRODUCTION

Spreadsheets such as Microsoft Excel are ubiquitous tools for ad-hoc data management and analysis across many domains—from finance to science—even among non-programmers [1]. It is estimated that over one-tenth of the world’s population uses spreadsheet software [2]. However, the growing volumes of data have pushed these tools to their limits. Although modern Excel supports up to approximately one million rows per worksheet [2], [3]—a capacity unimaginable in earlier versions—simply increasing this limit does not ensure scalable performance. In practice, spreadsheets often become unresponsive when handling only a few hundred thousand rows [1], [4], underscoring that while spreadsheets remain indispensable, they are inherently ill-suited for managing very large datasets compared to database systems. This tension between ease-of-use and scalability is the motivation for our research.

A particularly frustrating symptom of this limited scalability is *Excel file bloat*—the tendency of `.xlsx` files to become excessively large and slow. Users frequently encounter workbooks that swell in size far out of proportion to any actual data contained within them [5]. Such bloat is not just a storage

inconvenience; it also directly degrades performance, causing slow load/save times, heavy memory consumption, and delays in downstream data processing pipelines [6]. Understanding the causes of this bloat is therefore crucial. Prior reports and anecdotal evidence highlight several culprits. One common cause is excess formatting: Excel is notorious for retaining formatting in cells far beyond the used data range, silently inflating file size [7]. In effect, millions of “blank” cells may carry font, color, or border settings that contribute to the XML file, dramatically increasing the package size. Another cause is an overabundance of cell styles. Every unique combination of formatting (font, fill, border, etc.) is stored in a centralized style table; copying worksheets between workbooks or indiscriminate cell styling can accumulate tens of thousands of distinct styles. Excel allows up to 64,000 unique cell styles, and hitting this limit or even a fraction of it can bloat the `styles.xml` part of the file and trigger errors. [8]. In one documented case, Excel began warning the user that “this workbook has many duplicate styles which can slow performance,” prompting a cleanup of hundreds of extraneous style entries [9]. These examples underscore that Excel’s own features for flexibility (rich formatting, open XML storage) can turn into liabilities at scale.

The significance of the Excel bloat problem extends beyond file size annoyance—it poses broader scientific and technical challenges. First, it exemplifies the difficulties of managing large semi-structured data in a format not optimized for size or speed. The Office Open XML format (used by `.xlsx`) trades efficiency for interoperability, using verbose XML to represent spreadsheet content. When a workbook contains huge numbers of cells or lengthy text, the XML overhead (tags, attributes, and repetition) adds substantial bulk. A DOM parser, which builds an in-memory tree of the entire XML, can easily exhaust memory on multi-hundred-thousand-row sheets. For example, the memory consumption of a XML DOM parser (like `readxl`) can exceed ten times the size of the uncompressed worksheet (30 GB for 6,000,000 rows) [10]. Streaming SAX parser avoids high memory usage but suffer from high CPU overhead due to handling a flood of callback events [10].

Such scenarios reflect the mismatch between Excel’s data model and big-data processing needs. Processing such bloated

XML files is computationally expensive: parsing XML requires reading and interpreting deeply nested structures, which can strain I/O and memory. Indeed, standard XML parsing in memory (DOM) would load the entire spreadsheet into a tree, potentially consuming gigabytes of RAM for a bloated file. This is aggravated by Excel-specific XML components like the Shared Strings table (which holds all text entries) and the Styles table – both can grow very large and must be parsed or loaded before data can be accessed. As a result, reading a large .xlsx file with conventional libraries often becomes a bottleneck [11]. Second, the issue brings to light fundamental challenges in language theory and parsing. Excel’s file format—essentially a collection of XML documents—is a context-free language (Type-2 in the Chomsky hierarchy). Parsing such documents correctly requires managing recursive tag structures and dependencies (e.g., matching `<row>` and `</row>` tags or resolving references in shared string tables). It is mathematically impossible to accurately parse or validate XML using only regular expressions or simplistic line-by-line methods, since regular languages (Type-3) cannot handle arbitrary nested structures [12], [13]. Consequently, robust tools for processing .xlsx files at scale must employ proper XML parsing strategies rather than relying on ad-hoc techniques.

This paper situates the Excel bloat problem within the context of database research and proposes technical approaches to mitigate it. We begin by analysing the internal structure of Excel’s Office Open XML format to identify the root causes of file bloat, and review both related academic work and practitioner-oriented resources addressing bloat in Excel files. To ground this analysis, we examine manual guidelines as well as tools such as Microsoft Excel’s Performance Tab and the Excel Inquire Add-In, and trial versions of proprietary utilities including NXPowerLite Desktop, to assess their file size reduction behaviour.

After applying these manual guides and automatic tools to real-world Excel files, we perform a qualitative analysis of the XML code to examine the changes responsible for file size reduction. By comparing the visual integrity of the files before and after cleaning—and incrementally reverting modifications to pinpoint when visual errors disappeared—we gain insight into Excel’s internal file structure and determine which cleanup operations are safe. Building on this reverse-engineering process, we identify a set of safe modifications and implement them in our own prototype tool - Excel Shrink Analyzer [14] - using regular DOM-XML parsing for fast prototyping, to test these changes across a large number of files and validate the outcome by file size reduction and if the Spreadsheet still looks the same, to find the set of modifications that are safe and effective.

Following the prototype phase, we develop a tool applying the same modifications, but this time efficiently, using a state-machine-guided, regex-driven parser designed to clean bloated Excel files. This parser operates directly on the raw XML byte stream to overcome the limitations of conventional XML parsers when dealing with large, deeply nested structures. We evaluate the performance of this tool — referred to as Excel

Shrink [15] — on the same Dataset a series of Excel files by comparing the loading times of pre-cleaned files versus the original files when opened with standard libraries and Microsoft Excel itself.

The scope of this work is intentionally focused on Microsoft Excel and its Office Open XML representation. The observed bloat patterns arise primarily from interactive, user-driven editing practices—such as repeated formatting, copy-and-paste operations, and manual range manipulation—that are characteristic of Excel usage.

To guide our study, we pose the following research questions:

- 1) **(RQ1)** Are the root causes of Excel file bloat already well understood and documented in existing research, or is further empirical analysis needed to fully identify and quantify these causes?
- 2) **(RQ2)** Can a regex-driven, state-machine-guided parser overcome the typical limitations of regular expressions in processing structured XML data (e.g., prevent over-matching and preserve XML validity)?
- 3) **(RQ3)** Does precleaning bloated Excel files with such a method significantly improve overall loading time (*i.e.*, cleaning + loading) compared to loading the original file directly in Excel?

Through this exploration, we aim to bridge the gap between Excel’s convenience and the demands of very large-scale data management. Ultimately, our work aspires to improve the reliability and scalability of processing large Excel files by providing both developers and end users with actionable insights and practical tools. Developers gain a deeper understanding of the causes of file bloat and strategies for efficient cleanup, while Excel users receive a means to pre-clean files before further processing or to integrate Excel Shrink into automated workflows.

## II. BACKGROUND

### A. Excel’s Open XML Format

Excel workbooks (.xlsx) are stored in the Office Open XML format, which organizes spreadsheet data into multiple XML parts within a ZIP container. Notably, each worksheet is an XML file (sheet1.xml, sheet2.xml, etc.) containing the cell data (values, formulas) for that sheet [16]. In addition, there are global parts such as workbook.xml (overall workbook meta-data like the names of the sheet and their order), styles.xml (definitions of cell, row and column formats), and shared-Strings.xml (the table of reoccurring texts in the file). The Shared String Table is especially important for understanding Excel’s storage mechanism. It is a construct that contains one instance of each unique text string appearing in the workbook. Instead of writing repeated text values in every cell, Excel stores each distinct string once in sharedStrings.xml and then references it by index from the sheet files. This design aims to reduce file size and memory usage when many cells contain duplicate text (e.g., a column of repeating categorical values) [17], [18]. Each text cell in a sheet XML may look like `<c`

`r="A1" t="s"><v>42</v></c>`. This means that the cell at position A1 does not contain the number 42 itself. Instead, the attribute `t="s"` marks the cell’s content as a shared string reference — the `<v>` tag holds the index (here: 42) of the actual text stored in the file `sharedStrings.xml`. When Excel opens the worksheet, it looks up index 42 in that shared string table to retrieve and display the correct text in the cell. In scenarios with highly repetitive text, the shared string table yields smaller files.

Another central component is the Styles Part (`styles.xml`), which defines all cell formatting used in the workbook. Excel supports rich formatting options for cells—such as fonts, colors, number formats, and borders—and manages these through style records. Each style record represents a specific combination of formatting properties and can be shared among multiple cells, rows, or columns. Instead of duplicating formatting information, these elements simply reference the corresponding style record via the `s` attribute. [19].

### B. Prior Observations and Literature

Despite the prevalence of the Excel bloat problem in practice, there is scant academic literature that directly addresses it. To our knowledge, no peer-reviewed study thus far has focused on the internal inefficiencies of Excel’s XML format or the mitigation of bloated spreadsheets. This is consistent with the fact that the database research community has traditionally viewed spreadsheets as end-user tools rather than enterprise-scale data management systems. However, indirectly, several lines of work and industry reports touch on related issues. Microsoft’s official guidelines and support articles acknowledge problems such as the “Too many different cell formats” error (triggered by style overflow) [8] and provide best practices for reducing workbook size (e.g., removing unused styles, turning off pivot cache, etc.). The Excel user community (blogs, forums) is rife with case studies of bloated files and remedies. For example, Power [7] enumerates common reasons for oversized files and recommends using a ZIP tool to inspect which internal parts are largest - an approach that reveals whether data, formatting, or media are to blame. On Stack Overflow, Superuser and ExcelForum, numerous Q&A threads document instances of mysterious file growth, often resolved by cleaning formats or rebuilding the workbook from scratch [20]–[23]. These sources, while not formal research, provide valuable insight into the root causes of bloat and confirm that our problem formulation is grounded in reality. However the solutions they provide are often manual, some of them utilizing the Excel built in features, recently introduced features like the excel **Check Performance Tab** or the older Excel Inquire Add In, which are not suitable for automation pipelines and not scalable for large numbers of files. Yet the causes in the XML structure of the Excel file is not addressed in these sources and can only be identified by reverse engineering the cleanup tools or Microsoft excel itself by try and error removing parts of the XML file and checking if the file still looks the same.

In the academic sphere, there is growing recognition that spreadsheets and databases need not be disjoint worlds. Some

research efforts have looked at enhancing spreadsheets to handle larger datasets by leveraging database technology in the background. DataSpread [1] and Hillview [4] are two systems that retain an Excel-like interface but manage data in a relational database to improve scalability.

While these works are not about Excel’s file format per se, they underscore a broader point: when spreadsheets approach the size of databases, one must apply database-like solutions. Our work aligns with this philosophy by treating Excel bloat and processing as a data management problem. Instead of dismissing bloated spreadsheets as user error, we view them as an important issue in large-scale semi-structured data processing that warrants efficient techniques.

## III. RELATED WORK

A substantial body of research in top-tier venues (VLDB, SIGMOD, ICDE) has explored techniques to *improve performance* on large or complex workbooks. There have also been several efforts to reduce *Excel file size*, with various online resources discussing this topic. We briefly review key findings below, focusing on (i) methods for minimizing file size, (ii) approaches to accelerating computation and user interactions for large spreadsheets, (iii) optimizations for parsing the XML-based `.xlsx` format, and (iv) highlights of Aditya Parameswaran’s contributions in bridging spreadsheet and database technologies.

**Minimizing Excel File Size and Bloat.** Microsoft’s transition from the binary `.xls` format to `.xlsx` (Office Open XML) introduced compression and a shared string table to reduce repeated data [24], [25]. Further reduction is achieved in the `.xlsb` (binary) format, which stores data in a compact binary container at the cost of certain compatibility constraints [6], [26].

However, there is a lack of peer-reviewed studies specifically quantifying how “bloat” (invisible formatting, rows and column definitions outside of the visible content) affects Excel file size [27]. Instead, most advice on the subject comes from practitioners sharing their experiences. For example, Microsoft’s support article on cleaning excess cell formatting provides practical tips for reducing file size and improving performance [26]. Additionally, discussions on forums such as MrExcel illustrate that users have long noticed dramatic size reductions after clearing unused formatting or trimming the “used range” [28]. Popular Excel blogs from UpSlide [29], The Bricks [30] and Chandoo [31] further detail these methods based on accumulated user experience rather than formal experimentation.

Several proprietary tools have been developed to address Excel bloat automatically. Notable examples include UpSlide’s Automation Software, which offers a one-click “Smart Clean” feature [29]; NXPowerLite Desktop [32]; Sobolsoft’s Excel File Size Reduce Software [33]; and online services such as WeCompress [34]. Other tools, such as Kernel for Excel Repair [35], have also been mentioned in user communities for their file compression capabilities.

Microsoft has also developed two tools to assist users. One such tool is **Excel Inquire**, an add-in introduced with Excel 2013 that provides a range of auditing features—including Workbook Analysis, Relationship Diagrams, and the Clean Excess Cell Formatting command. O’Beirne notes that the *Clean Excess Cell Formatting* command removes excess formatting but does not indicate beforehand what will be cleared, nor does it report afterwards what was removed [36].

More recently, Microsoft introduced the **Excel Performance Tab**, a built-in feature designed to monitor workbook performance by identifying inefficiencies and offering cleaning recommendations. However, the Performance Tab has received negative user feedback; several users report that its forced “Check Performance” pop-up and the associated reformatting remove intentional formatting and disrupt workflows [37], [38] and also Excel users report that their pixel art is deleted [37] while the Microsoft Support Site even mentions a warning that pixel art will be removed by the tool [38].

In contrast, our paper provides a comprehensive scientific analysis of the root causes of Excel file bloat in XML-based workbooks. To address this gap, we introduce *Excel Shrink*—a command-line tool designed for automation pipelines. Unlike proprietary tools that are only accessible through a graphical user interface and sometimes remove too many elements (thereby altering the visual appearance) or too few (leaving the file size large), Excel Shrink selectively reduces file size by eliminating only those elements that do not affect the workbook’s visual integrity.

**Performance in Handling Large Spreadsheets.** Rahman et al. [39] conducted a comprehensive study comparing Excel, LibreOffice Calc, and Google Sheets on sizable datasets. They revealed that despite nominal row limits of one million, Excel’s GUI would often freeze with mere tens of thousands of rows or extended formula usage. Similar findings appear in other benchmarking efforts [2], highlighting the lack of indexing in typical spreadsheet software. Addressing this, “DataSpread” [1] (VLDB 2015) and “Hillview” [4] (PVLDB 2019) offload data management to database engines or distributed clusters, respectively, while retaining a spreadsheet-like interface. These architectures stretch the notion of a “spreadsheet” to hundreds of millions (or billions) of rows by introducing parallelization, indexing structures, or incremental computation. In a different line of work, Bendre et al. [40] proposed an *anti-freeze* model for formula recalculation, allowing asynchronous updates so the user interface does not lock while large spreadsheets recompute.

Systems such as *DataSpread* and *Hillview* improve scalability by re-architecting the spreadsheet runtime around database-style storage and computation. Our contribution is complementary: rather than replacing Excel’s execution model, we reduce the overhead of *representing* and *loading* real-world workbooks by eliminating redundant OOXML content that inflates I/O, memory pressure, and parsing costs in downstream tools.

**Efficient Excel XML Processing.** As modern Excel files (.xlsx) are essentially XML (SpreadsheetML) within a ZIP

container, *parsing* can become a bottleneck for large sheets. Classical DOM-based XML parsing can exhaust memory on multi-hundred-thousand-row spreadsheets, whereas SAX-based streaming, though more memory-friendly, suffers from high CPU overhead through numerous event callbacks [10]. A hybrid parser that combines DOM and SAX parsing, along with regular expressions for extracting the relevant file contents (e.g. `openxlsx` in R), can still be slow and memory-intensive under worst-case scenarios. By contrast, parsers relying exclusively on DOM parsing (e.g. `readxl` in R) may achieve faster runtime performance but often at the cost of higher memory usage. For instance, the memory consumption of `readxl` can exceed ten times the size of the uncompressed worksheet [10]. To address this, Henze et al. [10] introduced *SheetReader*, a specialized streaming parser that integrates decompression and XML scanning, and parallelizes chunk-level parsing. This yields dramatically lower memory footprints (often 40× less) and faster loads (3× or more) compared to standard libraries. Similar to this Excel Shrink also aims to reduce the memory footprint by streamwise processing of small chunks of the XML file and only keeping the relevant parts in memory.

*SheetReader* and related work accelerate *general-purpose extraction* from SpreadsheetML via streaming and parallel parsing. *Excel Shrink* instead performs a *semantics-preserving transformation* of SpreadsheetML: it uses streaming, state-guided pattern matching to *delete* redundant structures (e.g. empty styled cells, and out-of-range rows/columns) so that subsequent consumers (Excel itself and third-party libraries) face a substantially smaller parsing workload.

**Comparisons of Parsing Approaches and Tools.** These studies collectively show a trade-off between *DOM-based* simplicity (at the cost of high memory usage) and fully *SAX-based* pipelines (often incurring higher CPU overhead). Hybrid or domain-specific solutions—like regex-based extraction or streaming chunk processors that skip unneeded style markup—strike a middle ground [10], [41]. Meanwhile, broad tool evaluations (Microsoft Excel vs. Google Sheets vs. LibreOffice Calc) confirm that no mainstream spreadsheet program handles large data sets efficiently out of the box [39].

**Parameswaran’s Contributions: Bridging Spreadsheets and Databases.** Aditya Parameswaran and collaborators have researched integrating spreadsheets with database backends to achieve both usability and scalability [1], [42]. *DataSpread* preserves the familiar Excel-like interface while relying on a relational DBMS for storage and large-scale computations, enabling interactive performance even at billions of cells. This approach can be seen as bypassing Excel’s internal storage and recalculation limits by adopting database features (indexing, query optimization, concurrency control). In related work on “anti-freeze” recalculation, Parameswaran’s group explored asynchronous formula evaluation with incremental updates, preventing interface lock-ups for large or complex dependency graphs [40]. Additionally, they highlight practical tactics like saving spreadsheets as binary (.xlsb) instead of .xlsx to mitigate bloat, limiting volatile formulas (e.g. OFFSET),

and offloading heavy operations to databases where possible [6]. All these strategies offer evidence that a combination of advanced parsing, background recalculation, and robust data management techniques can significantly enhance performance for real-world Excel workloads.

We focus on a different leverage point: rather than replacing Excel’s execution model *Excel Shrink* targets the physical representation of `.xlsx` files. By eliminating redundant and visually inconsequential OOXML structures, our approach reduces I/O, memory pressure, and parsing cost while preserving full compatibility with the existing `.xlsx` ecosystem.

**Summary** Prior work improves spreadsheet scalability either by changing the execution architecture (e.g., database-backed spreadsheets) or by accelerating ingestion and parsing through specialised streaming designs. However, the community lacks systematic, reproducible evidence on *which* OOXML structures dominate file size and *which* transformations preserve visual integrity.

This paper fills that gap by (i) empirically characterising common sources of `.xlsx` bloat, and (ii) introducing a streaming transformation approach that reduces bloat while retaining the `.xlsx` format for compatibility with existing Excel-based and programmatic workflows.

#### IV. METHODOLOGY

*Excel Shrink* is a stream-oriented `.xlsx` cleaner designed to remove *structural bloat that is provably invisible to the user* while preserving all layout- and semantics-relevant content. By default, the system follows a conservative safety policy: only XML elements whose removal cannot affect visual appearance or formula evaluation are pruned. Operations that may alter semantics or layout are intentionally excluded unless explicitly enabled by the user.

**Whitespace handling policy.** Whitespace-only cells are treated as *potentially relevant* content. Although often unintentional, such cells may affect formula semantics (e.g., tests for non-emptiness or string concatenation) or serve layout purposes, such as preventing text from visually overflowing into adjacent cells. Therefore, *Excel Shrink* does *not* remove whitespace-only cells by default.

**System overview.** *Excel Shrink* operates directly on the bytestream of the zipped OOXML container. It parses `xl/workbook.xml` and per-sheet `xl/worksheets/sheet*.xml` streams using regular expressions instead of materialising full XML trees in memory, applies workbook-specific rewriting rules derived from `xl/styles.xml` and `xl/sharedStrings.xml`, and re-emits a compacted archive. **Edge cases and robustness.** A common concern with regex-based SpreadsheetML processing is whether user-entered text could accidentally match XML patterns. This cannot occur for valid Excel files: Excel escapes XML-like input. As a conservative safeguard, processing aborts upon encountering CDATA, which Excel itself treats as malformed. We further rely on SpreadsheetML’s mandated element order (violations are rejected by Excel), while remaining agnostic to attribute ordering: all regular

expressions are written to match attributes independent of order. Finally, prefixed/non-default namespaces (e.g., `x:worksheet`) are passed through unchanged to avoid data loss; such cases were only observed in machine-generated spreadsheets in our dataset and did not exhibit user-induced bloat.

**Formula semantics.** Removing *empty* cells, rows, or columns does not change formula semantics, even when these coordinates are referenced. *Excel Shrink* removes only their serialized representations from the worksheet XML; upon loading, Excel materializes missing cells as empty and evaluates references without introducing `#REF!` errors.

##### A. Inputs, Outputs, and Notation

**Inputs.** *Path* may be a single XLSX file, a directory, or an HTTPS/HTTP URL. Optional flags control recursion, multiprocessing, chunk size for streaming, sheet filters, and whether only the workbook is cleaned.

**Outputs.** A cleaned XLSX written either to a file path or into an output directory, preserving all unmodified parts of the original OOXML.

##### Core artefacts.

- *Workbook XML* (`xl/workbook.xml`): contains `<definedNames>` nodes.
- *Sheet XML* (`xl/worksheets/sheet*.xml`): contains `<cols>`, `<sheetData>`, `<mergeCells>`.
- *Styles* (`xl/styles.xml`): provides `<cellXfs>` with `fill/border/apply/protection` flags.
- *Shared Strings* (`xl/sharedStrings.xml`): enables identification of pure-whitespace string table entries.

**Workbook-specific patterns.** From `styles.xml` and `sharedStrings.xml` we derive:

- 1) sets of style indices with *fill* or *border* or with neither but *font-only*;
- 2) regex-like predicates for: (i) whitespace-only cells (for optional removal), (ii) consecutive empty cells/rows, (iii) attributes irrelevant for empty rows (*spans*, vendor *dyDescent*, etc.), and (iv) presence of *hidden* rows/columns.

##### B. Deriving Workbook-Specific Patterns

Our worksheet cleaning approach adopts a two-stage cleaning procedure based on a chunking strategy. In the initial pre-processing phase, individual chunks are analysed and cleaned using lightweight operations that do not require any global context from the worksheet. Some of these operations require knowledge of the workbook structure, others can be applied independently to each worksheet. Despite their simplicity, these early operations are effective in eliminating a large portion of file bloat from Excel spreadsheets.

These operations remove visually inconsequential cells and rows, i.e., elements that contain no content and no layout- or semantics-relevant styling. Cells are considered visually relevant if their associated style specifies background fills, borders, or protection attributes. Font-only styling (e.g., font family or size) is insufficient to classify a cell as visually relevant. Cells marked as protected, or explicitly exempted

from protection, are conservatively preserved, as protection often encodes user intent (e.g., preventing modification of form templates).

To identify removable cells and rows, Excel Shrink derives workbook-specific classification rules from `styles.xml` and `sharedStrings.xml`. From the extracted style definitions, we determine which style indices correspond to (i) visually empty formatting, (ii) fill- or border-bearing cells, and (iii) protected or protection-exempt cells. Shared strings are analysed to identify whitespace-only entries (for optional removal). Based on these sets, we compile regular expressions that match cells and rows that are safe to remove. The construction of these sets and the resulting patterns is summarised in Algorithm 1.

Let  $S_{fbp}$  denote the set of all *fill/border/applyProtection* style indices, i.e. those `xf` records where either the `fillId`, `borderId`, or `applyProtection` attribute is non-zero. For example, if the parsed style table (`styles.xml`) contains

$$\begin{aligned} XFS[0].fillId &= 0, & XFS[1].fillId &= 3, \\ XFS[2].borderId &= 2, & XFS[3].applyProtection &= 1, \end{aligned}$$

then  $S_{fbp} = \{1, 2, 3\}$ . These indices are joined into a single alternation string  $FBP = "1|2|3"$ , which is used to form the regular expression

$$p_{fbh} = "(?:s="(?:1|2|3)"|"hidden="1)"$$

matching a row attribute that either has a style with a set fill, border or `applyProtection` attribute, or a row attribute that is hidden used to find the last visually relevant row.

Analogously,  $S_{font}$  represents the set of *font-only* styles—those that apply no fill or border at all. For instance, if

$$\begin{aligned} XFS[4].fillId &= 0, & XFS[4].borderId &= 0, \\ XFS[5].fillId &= 0, & XFS[5].borderId &= 0, \end{aligned}$$

then  $S_{font} = \{4, 5\}$  is incorporated into the patterns

$$\begin{aligned} p_{ra} &= \backslash s * (\text{spans}="[^"]+" \\ &\quad | [^:]+:\text{dyDescent}="[^"]*" \\ &\quad | s="(?:4|5)" ) \\ p_e &= "(?:<c\b(?:?![>])*s="[>])*" \\ &\quad | [^>]*s="(?:4|5)" \\ &\quad | [^>]*>)+ \end{aligned}$$

which respectively match redundant row attributes and consecutive font-only empty cells.

Next,  $W$  is the set of *shared string indices* that correspond to whitespace-only values (e.g. cells containing a single space or tab). For example, if `sharedStrings.xml` contains

```
<si><t> </t></si>
<si><t> </t></si>
<si><t>hello</t></si>
```

then  $W = \{0, 1\}$  and  $WS = "0|1"$ .

These indices produce the regexes

$$p_{ws} = <v>(?:0|1)</v>$$

$$p_{wc} = (<c\b [^>]*t="s"[^>]*?)><v>(?:0|1)</v></c>$$

which match the corresponding whitespace value tags and their enclosing `<c>` elements.

All resulting expressions  $p_{ra}$ ,  $p_e$ ,  $p_{fbh}$ ,  $p_{ws}$ , and  $p_{wc}$  are compiled into the regex set  $R$ . Together with the mapping  $XFS$  from style IDs to `xf` records, they form a complete *pattern context* that drives the subsequent byte-stream cleaning phase. This allows the cleaner to efficiently detect and remove visually empty cells and rows purely through regular-expression matching—without ever parsing the XML structure.

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**Algorithm 1:** Consolidated construction: compute alternations and build dependent regexes in the same branch.

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```
Function BUILDWORKBOOKPATTERNS( $B_{stylesXml}$ ,  $B_{sharedStringsXml}$ ):
   $xf s \leftarrow \text{parseCellXfs}(B_{stylesXml})$ 
   $S_{fbp} \leftarrow \{i \mid xfs[i].fillId \neq 0 \vee xfs[i].borderId \neq 0 \vee xfs[i].applyProtection = 1\}$ 
   $S_{font} \leftarrow \{i \mid (xfs[i].fillId \in \{0, \text{None}\}) \wedge (xfs[i].borderId \in \{0, \text{None}\}) \wedge \neg xfs[i].applyFill \wedge \neg xfs[i].applyBorder\}$ 
   $W \leftarrow \text{INDICESOFWHITESPACESI}(B_{sharedStringsXml})$ 
   $XFS \leftarrow \{ "0" \mapsto xfs[0], "1" \mapsto xfs[1], \dots \}$ 
  if  $|S_{font}| > 0$  then
     $OFS \leftarrow \text{join}(S_{font}, |)$ 
     $p_{ra} \leftarrow '\backslash s * (\backslash spans = "[^"]+?" | ' +$ 
     $' \backslash b [^\backslash s :]*?:dyDescent="[^"]*" | ' +$ 
     $' \backslash bs = "(?:' + OFS + ')" )'$ 
     $p_e \leftarrow$ 
     $' (?:<c\b (?:?![>])*bs = "[^>]*" | [^>]*' +$ 
     $' \backslash bs = "(?:' + OFS + ')" [^>]*>)+'$ 
  else
     $p_{ra} \leftarrow '\backslash s * (\backslash spans = "[^"]+?" | ' +$ 
     $' \backslash b [^\backslash s :]*?:dyDescent="[^"]*" )'$ 
     $p_e \leftarrow ' (?:<c\b (?![>])*bs = "[^>]*>)+'$ 
  if  $|S_{fbp}| > 0$  then
     $FBP \leftarrow \text{join}(S_{fbp}, |)$ 
     $p_{fbph} \leftarrow$ 
     $' (?:s="(?:' + FBP + ')"|"hidden="1)"'$ 
  else
     $p_{fbph} \leftarrow ' "hidden="1"'$ 
  if  $|W| > 0$  then
     $WS \leftarrow \text{join}(W, |)$ 
     $p_{ws} \leftarrow '<v>(?:' + WS + ')</v>'$ 
     $p_{wc} \leftarrow$ 
     $' (<c\b [^>]*? \backslash bt = "s"[^>]*?)><v>(?:' +$ 
     $+ WS + ')</v></c>'$ 
  else
     $p_{ws} \leftarrow \text{None}$ 
     $p_{wc} \leftarrow \text{None}$ 
   $\mathcal{P} \leftarrow \{p_{ra}, p_e, p_{fbph}, p_{ws}, p_{wc}\}$  with COMPILE for all non-None
  return  $\langle \mathcal{P}, XFS \rangle$ 
```

---

These patterns parameterise all sheet and workbook transformations that follow, ensuring we preserve semantically relevant visual styling.

### C. Streaming Sheet Pre-processing

We implement our spreadsheet parser streaming-based, processing the XML data incrementally to minimize memory consumption and permit early discarding of unnecessary file

**Algorithm 2:** Streaming extraction of sheet components**Function**

```

PREPROCESSSHEETSTREAMWISE(sheetStream, P, chunkSize):
  state ← IN_WORKSHEET; b ← ε;
  BpreCols, Bcols, BpreSD, BsheetData, BpreMC, BmergeCells, Btail ←
  ε;
  T ← ∅; processed tags
  T ← ∅; processed tags
  Dcol ← newRangeMap(); column defs (map)
  mmax ← ⊥;
  O ←
  {<cols>, <sheetData>, <mergeCells>, </worksheet>};
  while state ≠ FOUND_END do
    b ← b || read(sheetStream, chunkSize);
    if state = IN_WORKSHEET then
      R ← O \ T;
      ⟨tag, itag, jtag⟩ ← nextTag(b, R);
      if tag ≠ ⊥ then
        if tag = <cols> then
          BpreCols ← BpreCols || b[0:itag]
          if tag = <sheetData> then
            BpreSD ← BpreSD || b[0:itag]
            if tag = <mergeCells> then
              BpreMC ← BpreMC || b[0:itag]
              b ← b[jtag:|b|];
              state ← nextState(tag);
              T ← T ∪ {tag};
        else if state = IN_COLS then
          ⟨c, b, state, Dcol⟩ ← CONSUMECOLS(b, P, Dcol)
          Bcols ← Bcols || c
        else if state = IN_SHEETDATA then
          ⟨c, b, state⟩ ← CONSUMESHEETDATA(b, P)
          BsheetData ← BsheetData || c
        else if state = IN_MERGECELLS then
          ⟨c, b, state, mmax⟩ ←
          CONSUMEMERGECELLS(b, mmax)
          BmergeCells ← BmergeCells || c
        else
          Btail ← Btail || b[0:|b|];
          b ← ε;
          break
  M ← {Dcol, mmax}
  return
  Preprocessed (BpreCols, Bcols, BpreSD, BsheetData,
  BpreMC, BmergeCells, Btail, M)

```

**Function** CONSUMESHEETDATA(b, P):

```

i ← find(b, </sheetData>)
if i ≥ 0 then return
  ⟨PREPROCESSSD(b[0:i], P), b[i:], IN_WORKSHEET⟩
k ← rfind(b, <row>)
if k ≥ 0 then return
  ⟨PREPROCESSSD(b[0:k], P), b[k:], IN_SHEETDATA⟩
return ⟨ε, b, IN_SHEETDATA⟩

```

sections. Below, we focus on how the parser transitions between states (i.e., how each state function decides to proceed), deferring lower-level details of in-state processing until later sections (Algorithm 2).

**States Overview** We define six main states in our parser:

- **BEFORE\_COLS:** The parser has not yet encountered the <cols> element. It searches for either <cols> or <sheetData>.
- **IN\_COLS:** The parser found <cols> and is processing column definitions.
- **BEFORE\_SHEET\_DATA:** The parser has finished reading

(or did not detect) the <cols> node and is searching for the first <sheetData> element.

- **IN\_SHEET\_DATA:** The parser is processing the main worksheet data (rows and cells).
- **AFTER\_SHEET\_DATA:** The parser has detected the closing </sheetData> and is collecting any subsequent chunks until the end of the file.
- **END\_OF\_FILE:** The parser has reached the end of the stream, and no further parsing is needed.

**In between nodes** In **BEFORE\_COLS**, **BEFORE\_SHEET\_DATA**, or **AFTER\_SHEET\_DATA**, the parser looks for tag boundaries (e.g., <cols>, <sheetData>) or the end of file. When it detects such a tag at position *i* in the buffer:

- 1) It *cuts* the buffer at *i*, splitting the buffer into two segments: (1) from the start to the end of the located tag, and (2) the bytes after that tag.
- 2) It returns the first segment as a chunk to add to the output, which the parser outputs or processes for the current state.
- 3) The second segment becomes the updated buffer, ensuring the XML node boundary is never split across segments.

If the parser fails to find any relevant tag or end of file, it returns no new output chunk, remains in the same state and leaves the buffer unmodified. This ensures that subsequent incoming data can be appended, eventually providing enough content to match the target boundary.

**In column definition node** In **IN\_COLS**, the parser searches for </cols> or any <col .../> elements. On detecting </cols>, it cuts the buffer right before the closing tag and transitions to **BEFORE\_SHEET\_DATA**. If it does not find the closing tag but locates complete <col .../> elements, those are consumed and removed from the buffer as a clean cut. If no match is found, the buffer remains unchanged, waiting for more data to complete the matching.

**Algorithm 3:** PreProcessSD — streaming prefilters for <sheetData> chunks

```

Data: phid ← compile(
  '(<row\b [^>]*?\bhidden="1"[^>]*+ +
  '(<:/>|>(:?(?!</row>|<v>)))+</row>))' +
  '(<:row\b [^>]*?\bhidden="1"[^>]*+ +
  '(<:/>|>(:?(?!</row>|<v>)))+</row>))'
)

```

**Function** PREPROCESSSD(b, P):

```

if has(b, hidden="1") then
  b ← sub(phid, "1", b)

if has(P.pws) ∧ has(P.pwc) then
  if match(P.pws, b) then
    b ← sub(P.pwc, "<1>", b)

if has(b, <c>) then
  b ← sub(P.pe, "<c>", b)

return b

```

**In sheet data node** While **IN\_SHEET\_DATA**, the parser similarly looks for </sheetData> or row boundaries <row ...>...</row> (or <row .../>). If </sheetData>

is found, the parser cuts the buffer at that tag and transitions to AFTER\_SHEET\_DATA. If no boundary is present yet, the buffer remains unaltered to combine with future data chunks.

**Ensuring “Clean Cuts” within the Streaming Buffer** Our parser processes XML data in a streaming fashion using a working buffer that preserves XML node boundaries, ensuring that no node is split across buffer cuts. State transitions are performed only after a complete and relevant XML tag has been identified.

If no such tag can be detected, the buffer is incrementally extended with subsequent input chunks without altering the parser state or consuming already buffered content. This process is repeated until either a valid cut point is identified or the end of the input stream is reached.

If the end of the stream is reached while the parser remains inside a nested XML node and no valid closing tag can be identified, parsing is aborted and the file is skipped. This condition indicates malformed or corrupted XML; skipping such files prevents partial processing and potential data loss.

**Buffer Updates and State Progress** By constraining parsing decisions to these well-defined cuts, the parser never splits an XML node across multiple buffers. Each function (1) cuts on the matching tag boundary and updates the parser state, or (2) leaves the buffer intact if no complete boundary is found, thus allowing the next chunk to complete it. This way the parser can process the XML incrementally without needing to store the entire document in memory and it is ensured that the regular expressions don’t fail due to incomplete XML nodes and also only the regular expressions that are relevant to the current section of the document are applied, increasing the performance in comparison to applying all regular expressions on the whole document.

The sub-routine `preProcessSheetData` (Algorithm 3) performs three linear passes over chunks: (i) optionally identify and collapse explicit whitespace-only cells via shared-string indices (disabled by default to preserve semantics and layout), (ii) remove consecutive empty cells (either styleless or font-only), and (iii) opportunistically remove runs of hidden rows while retaining the first sentinel when needed to preserve row height/visibility semantics.

Two of these transformations depend on workbook-specific patterns derived from `styles.xml` and `sharedStrings.xml`, as described in Section IV-B: (i) the removal of consecutive empty cells with no styling or font-only styling, and (ii) the optional detection and removal of explicit whitespace-only cells. In addition, we collapse consecutive hidden rows into a single sentinel row during pre-processing.

In Excel’s internal representation, rows may be marked as hidden using the attribute `hidden="1"`. Although such rows are visually collapsed in the spreadsheet, their explicit representation in the worksheet XML can extend unnecessarily over thousands or even millions of entries. In practice, we observed worksheets in which, beyond a certain point, all subsequent rows were marked as hidden. Excel renders all following rows as hidden once a single row with `hidden="1"` is

**Algorithm 4:** `SplitSheetDataRanges` — compute *cellsRange* and trailing *rowsRange*

---

**Function** `SPLITSHEETDATARANGES( $B_{\text{sheetData}}, \mathcal{P}$ ):`

```

 $i_{\text{lastC}} \leftarrow \text{rfind}(B_{\text{sheetData}}, "<c ")$ 
if  $i_{\text{lastC}} = -1$  then
   $i_{\text{cellsEnd}} \leftarrow -1$ 
else
   $j \leftarrow \text{find}(B_{\text{sheetData}}, "</row>", i_{\text{lastC}})$ 
  if  $j = -1$  then
    raise Error: "no </row> after last <c"
   $i_{\text{cellsEnd}} \leftarrow j + |"</row>"|$ 

 $i_{\text{visAttr}} \leftarrow i_{\text{cellsEnd}}$ 
 $m_{\text{last}} \leftarrow \perp$ 
foreach  $m \in \text{finditer}(\text{patterns.pfbph}, B_{\text{sheetData}}, \text{start} = i_{\text{cellsEnd}})$  do
   $m_{\text{last}} \leftarrow m$ 
if  $m_{\text{last}} \neq \perp$  then
   $i_{\text{visAttr}} \leftarrow \max(i_{\text{visAttr}}, \text{start}(m_{\text{last}}))$ 

 $j_{\text{close}} \leftarrow \text{find}(B_{\text{sheetData}}, "</row>", i_{\text{visAttr}})$ 
 $j_{\text{self}} \leftarrow \text{find}(B_{\text{sheetData}}, ">", i_{\text{visAttr}})$ 
if  $j_{\text{close}} \neq -1 \wedge j_{\text{self}} \neq -1$  then
  if  $j_{\text{close}} < j_{\text{self}}$  then
     $i_{\text{rowsEnd}} \leftarrow j_{\text{close}} + |"</row>"|$ 
  else
     $i_{\text{rowsEnd}} \leftarrow j_{\text{self}} + |">"|$ 
else if  $j_{\text{close}} = -1 \wedge j_{\text{self}} \neq -1$  then
   $i_{\text{rowsEnd}} \leftarrow j_{\text{self}} + |">"|$ 
else if  $j_{\text{close}} \neq -1 \wedge j_{\text{self}} = -1$  then
   $i_{\text{rowsEnd}} \leftarrow j_{\text{close}} + |"</row>"|$ 
else
  raise Error: "no </row> or /> after last anchor"

if  $i_{\text{cellsEnd}} = -1$  then
   $\text{cellsRange} \leftarrow \epsilon$ 
else
   $\text{cellsRange} \leftarrow B_{\text{sheetData}}[0 : i_{\text{cellsEnd}}]$ 

 $i_{\text{rowsStart}} \leftarrow \max(0, i_{\text{cellsEnd}})$ 
if  $i_{\text{rowsEnd}} = -1$  then
   $\text{rowsRange} \leftarrow \epsilon$ 
else
   $\text{rowsRange} \leftarrow B_{\text{sheetData}}[i_{\text{rowsStart}} : i_{\text{rowsEnd}}]$ 

return  $\langle \text{cellsRange}, \text{rowsRange} \rangle$ 

```

---

encountered, unless a later row explicitly overrides this setting. From a rendering perspective, a long sequence of hidden rows is therefore semantically equivalent to the presence of only the first hidden row.

To exploit this equivalence, Excel Shrink uses a regular expression to identify and collapse consecutive hidden rows into a single representative row. This transformation avoids expensive per-row checks and requires no document-wide context, making it well suited for chunk-level pre-processing. The resulting simplification significantly reduces document size before subsequent cleaning passes are applied. The construction and application of the corresponding regular expression are shown in Algorithm 3.

#### D. Post-processing of `<sheetData>`

After streaming prefilters and structural range extraction, the post-processing phase transforms the `<sheetData>` payload into a compact but layout-equivalent representation. The phase



---

**Algorithm 5:** `cleanRowsAndCellsInRange` — apply  $p_{\text{row}}$  with cell cleanup

---

```

Data:  $p_{\text{row}} \leftarrow \text{compile}($ 
  '<row\b ((?:[>]*\br = "([^\"]+)?' +
  ' (?:[>]*\bhidden = "(1)" )?' +
  ' (?:[>]*\bs = "([^\"]+)?' +
  ' (?:[>]*\bht = "([^\"]+)?' +
  ' [^>]*?) (?:/>|>(.*)</row>)'
 $p_{\text{cell}} \leftarrow \text{compile}($ 
  '<c\b ((?:[>]*\br = "[A-Z+]) (\d +) )?' +
  ' (?:[>]*\bs = "(\d +) )? [^>]*?) />'
 $\text{Function CLEANROWSANDCELLSINRANGE}(B_{\text{row}}, \mathcal{P}, \mathcal{M}, XFS):$ 
   $\text{return sub}(B_{\text{row}}, p_{\text{row}}, \text{REPLACEROW}(\cdot, \mathcal{P}, \mathcal{M}, XFS))$ 

Function REPLACEROW}(m, \mathcal{P}, \mathcal{M}, XFS):
   $(\text{rowAttr}, r, \text{hidden}, sId, \text{ht}, \text{content}) \leftarrow \text{groups}(m)$ 
   $\text{rowStyle} \leftarrow (sId \neq \perp) ? \text{xfs}[sId] : \perp$ 
   $\text{hasHeight} \leftarrow (\text{ht} \neq \perp), \text{wasEmpty} \leftarrow (\text{content} = \epsilon)$ 
   $\mathcal{M}.\text{rowAttributes}[r] \leftarrow (\text{rowAttr}, r, (\text{hidden} = 1), \text{rowStyle}, \text{hasHeight}, \text{wasEmpty})$ 
  if  $\neg \text{wasEmpty}$  then
     $\text{content} \leftarrow \text{CleanCells}(\text{content}, \text{patterns}, \mathcal{M}, \text{xfs})$ 
   $\text{isEmpty} \leftarrow (\text{content} = \epsilon),$ 
   $\text{becameEmpty} \leftarrow (\neg \text{wasEmpty} \wedge \text{isEmpty})$ 
  if  $\text{hidden} = 1$  then
    if  $\text{previousRowWasAlsoHidden}(\mathcal{M})$  then
       $\text{return ""}$ 
    else
       $\text{return '<row' + clean}(\text{rowAttr}) + '/>'$ 
  if  $\text{isEmpty}$  then
    if  $(\text{rowStyle} = \perp \vee \neg \text{rowStyle}.\text{hasFillOrBorder}) \wedge \neg \text{hasHeight}$  then
       $\text{return ""}$ 
    if  $\text{becameEmpty}$  then
       $\text{rowAttr} \leftarrow \text{ADDCUSTOMHEIGHTIFABSENT}(\text{rowAttr})$ 
       $\text{return '<row' + clean}(\text{rowAttr}) + '/>'$ 
  return
    '<row' + rowAttr + '>' + content + '</row>'

Function CLEANCELLS}(rowBytes, \mathcal{P}, \mathcal{M}, XFS):
  if  $\text{rowBytes} = \epsilon$  then
     $\text{return rowBytes}$ 
   $\text{return sub}(rowBytes, p_{\text{cell}}, \text{ReplaceCell}(\cdot, \mathcal{M}, XFS))$ 

Function REPLACECELL}(m, \mathcal{M}, XFS):
   $(\text{cellAttr}, \text{col}, \text{row}, sId) \leftarrow \text{groups}(m)$ 
   $\text{style} \leftarrow (sId \neq \perp) ? \text{XFS}[sId] : \perp$ 
   $\text{coord} \leftarrow (\text{col}, \text{row})$ 
  if  $\text{CELLISHIDDEN}(\text{coord}, \mathcal{M})$  then
     $\text{return ""}$ 
  if  $\text{style} = \perp$  then
     $\text{return ""}$ 
  if  $\text{style}.\text{hasFillOrBorder}$  then
     $\text{return '<c' + cellAttr + '/>'}$ 
  if  $\neg \text{style}.\text{hasApplyFillOrBorder}$  then
     $\text{return ""}$ 
  if  $\text{CELLOVERRIDESFILLORBORDER}(\text{style}, \text{coord}, \mathcal{M})$  then
     $\text{return '<c' + cellAttr + '/>'}$ 
  else
     $\text{return ""}$ 

```

---

comprises two linear passes that operate on disjoint byte slices returned by the splitter of Alg. 4: (i) a *rows+cells* sweep over the prefix slice that still contains cells (*cellsRange*), and (ii) a *trailing empty-rows* sweep over the suffix slice that no longer contains any cells (*rowsRange*). Both passes are guided by workbook-specific patterns (See IV-B) and update a small, per-sheet metrics state  $\mathcal{M}$  used by later steps (merge-column normalization and diagnostics).

### Range splitting

Alg. 4 identifies the last row that still contains a cell by scanning backwards for a `<c` token, then advancing to the end of that row. From that anchor it scans forward for the last *visibly relevant attribute*—either a fill/border/protection-bearing style (`s="FBP"`) or `hidden="1"`—by iterating a precompiled pattern  $p_{\text{fbph}}$ . The earliest of the next `</row>` and `/>` delimiters after this anchor marks the end of the trailing visible-rows region. The function returns two slices: the *cells\_range* from the beginning up to and including the last cell-bearing row, and the *rows\_range* from there up to the last visibly relevant row. This design avoids unnecessary checks, as all content beyond the last relevant row is guaranteed to be empty and can therefore be removed without further inspection, whereas the intermediate range (between the last cell-bearing and the last relevant row) is examined only for rows that are hidden or have an explicitly defined height.

**Row+cell sweep on the cells prefix** The first pass (Alg. 5) applies a row-level regex  $p_{\text{row}}$  to the *cellsRange*. Each row match is handed to *ReplaceRow*, which (i) records row attributes for later hidden/override checks, (ii) cleans the inner cells via *CleanCells* if present, and (iii) decides to delete the row, keep it as self-closing (empty but layout-relevant), or keep it with (cleaned) content. The cell policy is encapsulated in *ReplaceCell*: cells in hidden row/column, without style, or without visible effects are dropped; cells with fills/borders or with effective overrides are kept as self-closing. All actions are appended to  $\mathcal{M}$  for downstream use.

**Trailing empty-rows sweep** The second pass (Alg. 6) runs over the *rowsRange* using  $p_{\text{rowE}}$ . *ReplaceEmptyRow* preserves the first row of any hidden-run and any style-bearing empty row (to retain layout height/borders), while deleting layout-neutral empty rows. This pass never touches cells (by construction there are none in the suffix), which makes it both simpler and faster. Also  $p_{\text{rowE}}$  is slightly simpler and hence faster than  $p_{\text{row}}$  as it omits the cell content group.

**Metrics and column merging** The rows+cells pass records kept cell coordinates and row attributes in  $\mathcal{M}$ ; combined with merge-cell bounds and visible column definitions, those are then used to merge extraneous `<col>` ranges (See IV-E).

**Cell policy (summary)** A cell is kept (as self-closing) iff it has a visible effect on layout: it is not in a hidden row or column, it carries a cell style, and it either applies a fill or border or explicitly overrides an inherited *apply* style from its row/column. Otherwise the cell is deleted.

---

**Algorithm 6:** CleanEmptyRowsFromAfterCellRange — apply  $p_{rowE}$ 

---

```
Data:  $p_{rowE} \leftarrow \text{compile}$ 
'<row\b ((?:[>]*\br ="(["]+?)")?' +
' (?:[>]*\bhidden ="(1)"?)?' +
' (?:[>]*\bs ="(["]+?)")?' +
' [^>]*? ) (?:/>|></row>)'

Function CLEANEMPTYROWSFROMAFTERCELLRANGE( $x, \mathcal{P}, \mathcal{M}, XFS$ ):
   $\text{return sub}(x, p_{rowE}, \text{ReplaceEmptyRow}(\cdot, \mathcal{M}, XFS))$ 

Function REPLACEEMPTYROW( $m, \mathcal{M}, XFS$ ):
  ( $rowAttr, r, hidden, sId$ )  $\leftarrow \text{groups}(m)$ 
   $rowStyle \leftarrow (sId \neq \perp) ? xfs[sId] : \perp$ 
   $\mathcal{M}.rowAttributes[r] \leftarrow (rowAttr, r, (hidden =$ 
     $1), rowStyle, hasHeight = false, isEmpty = true)$ 
  if  $hidden = 1$  then
    if  $\text{previousRowWasAlsoHidden}(\mathcal{M})$  then
       $\text{return ""}$ 
    else
       $\text{return '<row' + clean(rowAttr) + '/>'}$ 
  if ( $rowStyle = \perp \vee \neg \text{hasFillOrBorder}(rowStyle)$ ) then
     $\text{return ""}$ 
  else
     $\text{return '<row' + clean(rowAttr) + '/>'}$ 
```

---

### E. Column Range Compaction

We compact  $\langle cols \rangle$  by merging “extraneous” trailing definitions that exceed the rightmost visible column inferred from (i) kept cells, (ii)  $\langle mergeCells \rangle$ , and (iii) last visible column definition (hidden or with fill/border). If the first extraneous range precedes others, we widen it to the last, replacing the suffix.

### F. Multi-process Execution

Our multi-process execution model relies on a process pool that distributes workbook and per-sheet cleaning tasks across multiple CPU cores. This level of concurrency is only feasible because *Excel Shrink* operates directly on the raw byte stream using regular expressions, rather than parsing the XML tree structure of each sheet.

If we instead used XML parsing—as in the earlier *Excel Shrink Analyzer* prototype—the memory footprint would already be saturated in single-process mode, with multi-gigabyte workbooks easily exhausting 8–16 GB of RAM on a typical consumer machine. Running 2, 4, or even 12 XML-based cleaning threads concurrently would therefore be impossible. By contrast, our byte-stream approach keeps memory consumption minimal and enables parallel execution even for extremely large workbooks.

Beyond memory efficiency, we further optimised scheduling. When a file with multiple sheets (e.g. 13) is being cleaned, up to 12 sheets can be processed in parallel. As soon as one process finishes its assigned sheet, it immediately fetches the next available task—either another sheet within the same workbook or, if all sheets are already processed, the next workbook in the queue. This dynamic scheduling ensures that no CPU core remains idle, except when the final workbook is being processed.

## V. EVALUATION

We evaluated *Excel Shrink* on 1,183 real-world Excel files from the German Federal Statistical Office (*Destatis*) [43], [44] and 8,643 real-world Excel files from *data.gov* [45]. For each workbook, we verified value-level equivalence before and after cleaning by recalculating all formulas and comparing the resulting cell values. All values were identical, except for cells containing volatile formulas (e.g.  $\text{=RAND}()$ ,  $\text{=NOW}()$ , or  $\text{=CELL}(\text{"filename"})$ ), whose semantics explicitly permit differing results across recalculations [46], [47].

Our primary goal was to quantify the extent of data reduction achieved by *Excel Shrink* under its default, conservative configuration. We therefore focus on metrics such as overall file size reduction, removal of empty cells, deletion of rows outside the range of actual data values, and elimination of unnecessary column definitions.

Figure 1 presents the distribution of reduction rates (in percentage) for individual worksheets. To disentangle the contribution of different bloat sources, these measurements were obtained using *Excel Shrink Analyzer*, which enables optional transformations (including whitespace removal) for diagnostic purposes. The chart shows that some worksheets experienced reductions approaching 100% in file size. In particular, one worksheet exceeded a 99.99% reduction, 16 worksheets surpassed 99.9%, and 210 worksheets achieved reductions above 99%. Across all worksheets, the average reduction rate was 25%. Most of these gains were realised by removing rows that lay outside the range of actual data values.

We further observed that optional whitespace removal yields only comparatively small additional size reductions, reinforcing the decision to preserve such cells by default in *Excel Shrink* given their limited contribution to bloat and their potential impact on visual appearance or formula evaluation.

To further assess the performance improvements enabled by *Excel Shrink*, we selected two particularly large workbooks from our *Destatis* and *data.gov* datasets [48], [49]. In their original compressed *Excel* (.xlsx) form, the files measured approximately 116 MB and 85 MB, respectively, and expanded by nearly an order of magnitude when uncompressed due to extensive sparsity and redundant formatting.

Figure 2 summarises runtime, file size, and peak memory consumption before and after applying *Excel Shrink*. Each file was processed using `openpyxl`, `pandas`, `Microsoft Excel`, `R openxlsx`, `R readxl+ writexl`, and our *Excel Shrink* pipeline. For *Excel Shrink*, we additionally report a single-core execution and a variant with `sheetdata` splitting disabled to isolate the impact of key algorithmic design choices.

a) *Loading and Saving Times*: After shrinking, loading and saving times decrease consistently across all libraries, often by more than an order of magnitude. The default multi-core configuration of *Excel Shrink* completes a full clean-and-save cycle in only a few seconds, while the single-core variant is slower but remains competitive, demonstrating effective parallel scalability without relying on excessive per-core

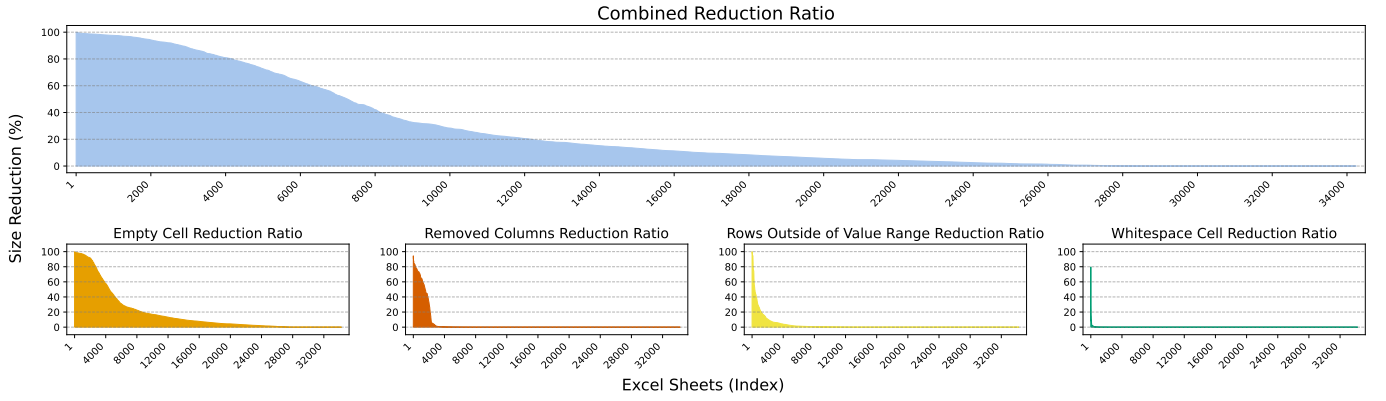


Fig. 1: Distribution of Size Reduction Percentages for Excel Sheets

memory. Disabling sheetdata splitting significantly increases runtime, as large sheet XML streams must be processed more exhaustively without early truncation (Algorithm 4).

b) *File Size Reduction*: Both workbooks shrink from approximately 116 MB and 85 MB to 3.1 MB and 4.9 MB, corresponding to reductions of about 97.4% and 94.2%, respectively. These reductions are independent of parallel execution and result solely from the removal of visually irrelevant XML content.

c) *Memory Consumption*: For the original workbooks, peak memory usage is substantial: `openpyxl` exceeds 8 GB on the larger file, while R-based libraries peak around 2–3 GB. After shrinking, peak memory usage drops below 1 GB across all tools. The single-core *Excel Shrink* variant exhibits particularly low memory consumption, highlighting the streaming nature of the core algorithms. In contrast, disabling sheetdata splitting increases peak memory usage, as larger XML fragments must be retained for analysis.

Overall, these results show that *Excel Shrink* achieves substantial file size reductions while enabling faster and more memory-efficient processing across spreadsheet ecosystems.

#### A. Experimental Setup

To evaluate the behavior of *Excel* regarding how the files are interpreted and saved, as well as to test the behavior of the *Add-in* and the new *Check Performance* feature, we tested with two distinct versions of *Excel* from *Microsoft Office*: *Microsoft Office LTSC Professional Plus 2021* and *Microsoft Excel for Microsoft 365 MSO (Version 2409 Build 16.0.18025.20030) 64-bit*. These versions were selected to evaluate compatibility and performance across different Excel environments. Our experiments were conducted on a laptop equipped with an AMD Ryzen 5 Pro 7535U processor (6 cores, 12 threads, 2.9 GHz base clock) and 32 GB of RAM.

#### B. Comparison with Other Tools for Reducing Bloat

1) *Excel Inquire Add-in*: The *Excel Inquire* Add-in, available only in *Office Professional Plus* and *Microsoft 365 Apps for Enterprise*, can remove some redundant content [50].

*Excel Inquire* can delete empty cell, column, and row definitions. Yet, in certain cases, a significant number of unnecessary elements remain. For example, consider a worksheet serving as a cover sheet, where only the first 77 rows are used and the remaining rows (78 to 65,533 (the old limit of row in Excel files before it was increased to 1,048,576)) are hidden. Despite being hidden, all these rows and cells are defined in the XML structure. When applying the *Excel Inquire* Add-in to such worksheets, these hidden rows and cells remain. *Excel Shrink* successfully removed those cells.

2) *Excel Check Performance*: The *Check Performance* feature, introduced in *Excel* for the Web and recent desktop versions, can achieve substantial file size reductions [51], [52]. However, it often removes cells and rows that influence the spreadsheet’s layout and formatting, thereby altering its visual appearance. This is unacceptable in contexts where maintaining the original look and structure of the spreadsheet is essential.

Figure 3 illustrates how *Check Performance* modifies a cover sheet by removing empty rows. Such modifications become problematic when these empty rows serve a functional purpose, such as aligning the layout for printing. In the example shown, the cells responsible for providing space on the cover sheet (Figure 3a) are removed, as shown in Figure 3b [53]. *Excel Shrink* preserves this layout.

## VI. FUTURE WORK

While *Excel Shrink* parses `styles.xml` and `sharedStrings.xml` for contextual correctness, it does not yet remove unused style definitions or unreferenced shared strings. Safely eliminating such entries—while preserving rendering and recalculation semantics—requires global reference tracking and additional validation, and is left for future work.

## VII. CONCLUSION

We have presented *Excel Shrink*, a tool that significantly reduces the file sizes of *Excel* sheets by eliminating unnecessary content. Our analysis identified common practices that lead to bloated *Excel* files and demonstrated how targeted removal

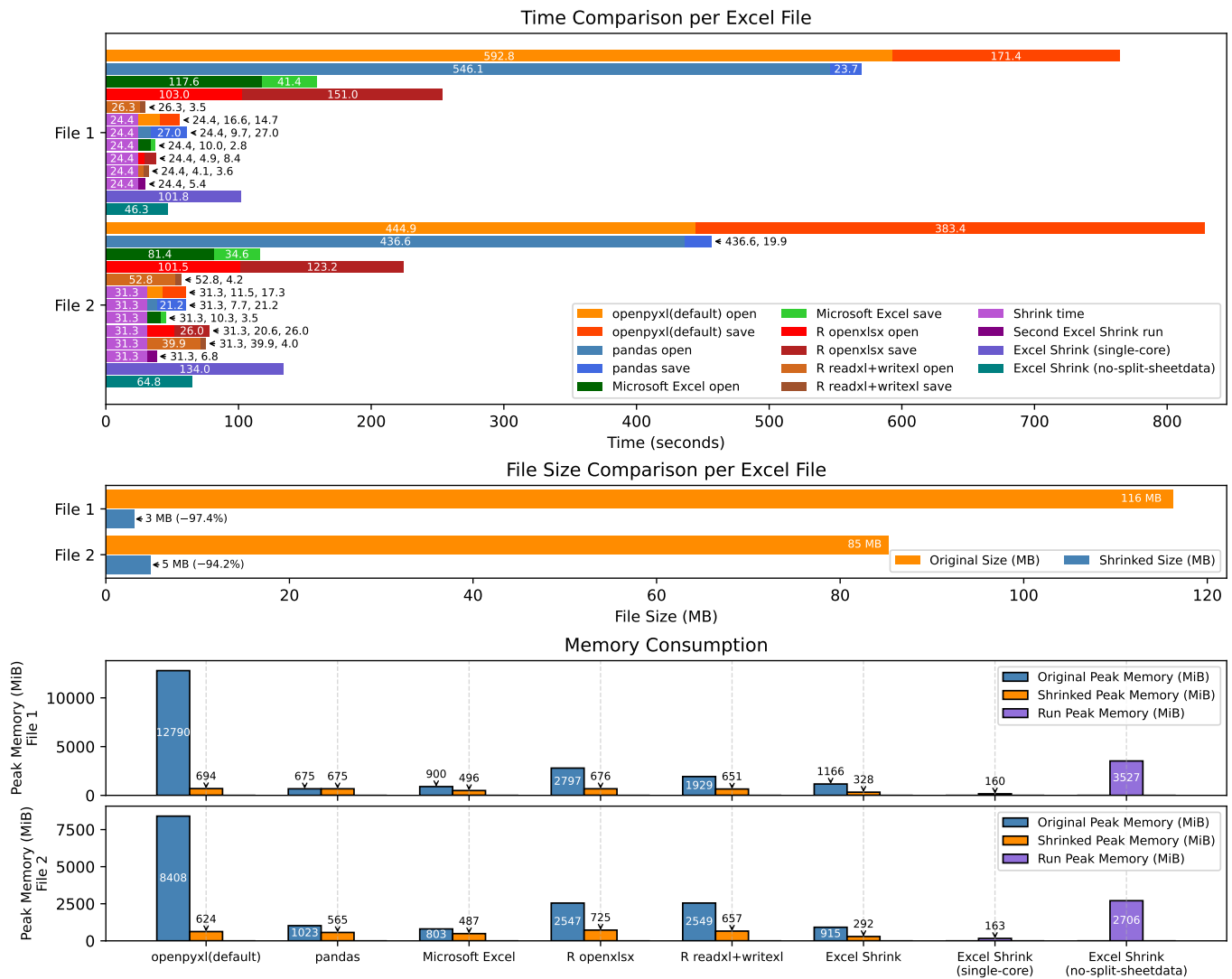


Fig. 2: Loading Times and Size Reduction for the largest *Destatis* and *data.gov* Files Before and After applying *Excel Shrink*

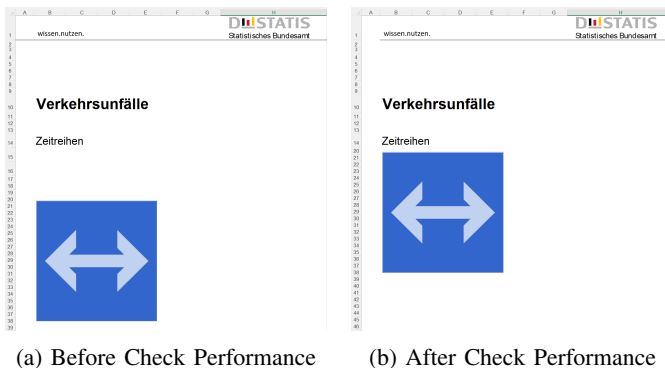


Fig. 3: Comparison of a Cover Sheet Before and After Applying Check Performance

of redundant data can achieve reductions of up to 99.99% in real-world Excel files without altering the spreadsheets'

appearance while outperforming existing solutions in both file size reduction, preservation of formatting and performance.

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